

RATIO & PROPORTION

Type-1

1. Rs.33,630 are divided among A, B and C in such a manner that the ratio of the amount of A to that of B is 3 : 7 and the ratio of the amount of B to that of C is 6 : 5. The amount of money received by B is

(1) Rs.14,868 (2) Rs.16,257 (3) Rs.13,290 (4) Rs.12,390

2. A, B and C are batsmen. The ratio of the runs scored by them in a certain match are given below $A : B = 5 : 3$ and

$B : C = 4 : 5$. In all they scored 564 runs. The number of runs scored by B is:

(1) 124 (2) 104 (3) 114 (4) 144

3. A person distributes his pens among four friends A, B, C, D in the ratio $1/3 : 1/4 : 1/5 : 1/6$. What is the minimum number of pens that the person should have?

(1) 57 (2) 65 (3) 75 (4) 45

4. If 177 is divided into 3 parts in the ratio $1/2 : 2/3 : 4/5$, then the second part is

(1) 75 (2) 45 (3) 72 (4) 60

Type-2

5. The incomes of 'A' and 'B' are in the ratio 5:3 and expenses are in the ratio 3:1. If each saves Rs.1000, what is B's expenditure.

1) Rs.2000 2) Rs.1500 3) Rs.500 4) Rs.1000

6. The Income of 'A' and 'B' are in the ratio 7:5 and expenses are in the ratio 5:3. If each saves Rs.2000, then what is their income?

1) Rs.12000 2) Rs.2000 3) Rs.800 4) None

7. A and B have monthly incomes in the ratio 5 : 6 and monthly expenditures in the ratio 3 : 4. If they save Rs.1800 and Rs.1600 respectively, find the monthly income of B :

(1) Rs.3400 (2) Rs.2700 (3) Rs.1720 (4) Rs.7200

8. The ratio of income of two persons is 5 : 3 and that of their expenditures is 9 : 5. Find the income of each person, if they save Rs.1,300 and Rs.900 respectively.

(1) 4,000, 2,400 (2) 3,000, 1,800 (3) 5,000, 3,000 (4) 4,500, 2,700

9. The income of A, B and C are in the ratio 3 : 7 : 4 and their expenses in the ratio 4 : 3 : 5. If A saves Rs.300 out of an income of Rs.2,400, the savings of B and C are :

(1) 4025 & 575 (2) 1575 & 2625 (3) 2750 & 1525 (4) 3725 & 1525

10. The income of A, B and C are in the ratio 7 : 9 : 12 and their spending are in the ratio 8 : 9 : 15. If A saves $1/4$ th of his income, then the savings of A, B and C are in the ratio of

(1) 56 : 99 : 69 (2) 69 : 56 : 99 (3) 99 : 56 : 69 (4) 99 : 69 : 56

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Type-3

11.If two numbers are in the ratio 2 : 3 and the ratio becomes 3 : 4 when 8 is added to both the numbers, then the sum of the two numbers is

(1) 10 (2) 80 (3) 40 (4) 100

12. Two numbers are in the ratio 3 : 5. If 9 is subtracted from each, then they are in the ratio 12 : 23. Find the smaller number.

(1) 27 (2) 33 (3) 49 (4) 55

13. In a regiment the ratio between the number of officers to soldiers was 3 : 31 before battle. In a battle 6 officers and 22 soldiers were killed and the ratio became 1 : 13, the number of officers in the regiment before battle was

(1) 31 (2) 38 (3) 21 (4) 28

14.The students in three classes are in the ratio 2 : 3 : 5. If 20 students are increased in each class, the ratio changes to 4 : 5 : 7. Originally the total number of students was :

(1) 50 (2) 90 (3) 100 (4) 150

15.The students in three classes are in the ratio 2 : 3 : 5. If 40 students are increased in each class, the ratio changes to 4 : 5 : 7.Originally, the total number of student was :

(1) 100 (2) 180 (3) 200 (4) 400

Type-4

16. Zinc and copper are in the ratio of 5 : 3 in 200 gm of an alloy. How much grams of copper be added to make the ratio as 3 : 5?

(1) $133\frac{1}{3}$ (2) $\frac{1}{200}$ (3) 72 (4) 66

17. In 30 liters mixture of acid, the ratio of acid and water is 2 : 3 . What amount of water should be added to the mixture so that the ratio of acid and water becomes 2 : 5 ?

(1) 10 litres (2) 15 litres (3) 18 litres (4) 12 litres

Type-5

18. One man adds 3 liters of water to 7 liters of milk and another 4 liters of water to 11 liters of milk. What is the ratio of strengths of milk in the two mixtures?

1) 21:22 2) 22:21 3) 7:3 4) 3:7

19. One man adds 6 liters of water to 4 liters of milk and another 8 liters of water to 7 liters of milk. What is the ratio of strengths of water in the two mixtures?

1) 10:11 2) 8:9 3) 9:8 4) 11:15

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Type-6

20. Two vessels contain milk and water in the ratio 3:4 and 5:2. If these two are mixed together then what is the ratio of milk and water?

1) 3:4 2) 4:3 3) 7:3 4) 3:7

21. The milk and water in two vessels A and B are in the ratio 4 : 3 and 5 : 2 respectively. In what ratio, the liquids in both the vessels be mixed to obtain a new mixture in vessel C

(1) 9 : 5 (2) 5 : 2 (3) 3 : 11 (4) 1 : 2

22. Two vessels contain milk and water in the ratio 4:3 and 3:2, if these two are mixed together then what is the ratio of milk and water?

1) 41:29 2) 29:41 3) 7:5 4) 5:7

23. Two vessels contain spirit and alcohol in the ratio 3:2 and 2:1, if these two are mixed together then what is the ratio of milk Spirit and Alcohol?

1) 11:19 2) 7:5 3) 5:7 4) 19:11

24. There are three containers of equal capacity. The ratio of Sulphuric acid to water in the first container is 3 : 2, that in the second container is 7 : 3 and in the third container it is 11 : 4. If all the liquids are mixed together, then the ratio of Sulphuric acid to water in the mixture will be :

(1) 61 : 29 (2) 61 : 28 (3) 60 : 29 (4) 59 : 292

25. Three utensils contain equal quantity of mixtures of milk and water in the ratio 6 : 1, 5 : 2 and 3 : 1 respectively. If all the solutions are mixed together, the ratio of milk and water in the final mixture is

(1) 65 : 28 (2) 65 : 19 (3) 19 : 65 (4) 19 : 28

Type-7

26. Two vessels contain spirit and alcohol in the ratio 3:1 and 2:3, if these two are mixed in the ratio 4:3 then what is the ratio of spirit and alcohol?

1) 3:2 2) 2:3 3) 5:2 4) 3:5

27. Three containers whose volumes are in the ratio of 2 : 3 : 4 are full of mixture of spirit and water. In the 1st container, the ratio of spirit and water is 4 : 1, in the 2nd container the ratio is 11 : 4 and in the 3rd container ratio is 7 : 3. All the three mixtures are mixed in a big container. The ratio of spirit and water in the resultant mixture is :

(1) 4 : 9 (2) 11 : 4 (3) 5 : 10 (4) 9 : 5

28. There are three bottles of mixture of syrup and water of ratios 2 : 3, 3 : 4 and 7 : 5. 10 liters of the first and 21 liters of the second bottles are taken. How much quantity from third bottle is to be taken so that final mixture from three bottles will be of ratios 1 : 1.

(1) 25 litres (2) 20 litres (3) 35 litres (4) 30 litres

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Type-8

39. A container contains 100 L of milk. From this container 10 L of milk was taken out and replaced by water. This process was further repeated three times. How much milk does the container have now?

1) 72.9 L 2) 65.61 L 3) 34.39 L 4) 81 L

30. A Can contains a mixture of two liquids A and B in the ratio 7:5. When 9 liters of mixture are drained off and the Can is filled with B, the ratio of A and B becomes 7:9. How many liters of liquid A were contained by the Can initially?

(1) 10 litres (2) 20 litres (3) 21 litres (4) 25 litres

31. A container contains 60 Lit of milk. From this container 6 Lit of milk was taken out and replaced by water. This process was repeated further two times. The amount of milk left in the container is

(1) 34.24 Lit (2) 39.64 Lit (3) 43.74 Lit (4) 47.6 Lit

Type-9

32. A box contains 1-rupee, 50-paise and 25-paise coins in the ratio 8 : 5 : 3. If the total amount of money in the box is Rs.112.50, the number of 50-paise coins is

(1) 80 (2) 50 (3) 30 (4) 42

33. A bag contains Rs.90 in coins of denominations of 50 paise, 25 paise and 10 paise. If coins of 50 paise, 25 paise and 10 paise are in the ratio 2 : 3 : 5, then the number of 25 paise coins in the bag is

(1) 80 (2) 120 (3) 100 (4) 135

34. A box contains 420 coins in rupee, 50 paise and 20 paise coins. The ratio of their rupee values being 13 : 11 : 7. The number of 50 paise coins is

(1) 42 (2) 78 (3) 66 (4) 132

35. A box contains 280 coins of one rupee, 50-paise and 25-paise. The values of each kind of the coins are in the ratio of 8 : 4 : 3. Then the number of 50-paise coins is

(1) 70 (2) 60 (3) 80 (4) 90

Type-10

36. A cat pursues a rat and takes 4 leaps for every 5 leaps of the rat, but 3 leaps of the cat are equal to 4 leaps of the rat. Compare the rates of the cat and the rat?

1) 5:4 2) 15:16 3) 16:15 4) 10:9

37. A wolf pursues a dog and takes 5 leaps of every 6 leaps of the dog, but 4 leaps of the wolf are equal to 5 leaps of the dog. Compare the rates of the dog and wolf?

1) 24:25 2) 25:24 3) 16:15 4) 15:16

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Type-11

38. Divide Rs.1320 among 7 men 11 women and 5 boys so that each woman may have 3 times as much as a boy, and a man as much as a woman and a boy together. Find how much each person receives?

1) Rs.80,60,20 2) Rs.80,20,60 3) Rs.20,60,80 4) Rs.60,20,80

39. An amount of Rs.375 was divided among A, B and C. If each of them had received Rs.25 less, their share would have been in the ratio 1:3:2. The money received by C was

1) Rs.100 2) Rs.75 3) Rs.175 4) Rs.125

40. The ratio of the first and second class train fares between two stations is 3 : 1 and that of the numbers of passengers travelling between the two stations by first and second classes is 1 : 50. If on a particular day, 1, 325 are collected from passengers travelling between the two stations, then the amount collected from the second class passengers is

(1) 1,250 (2) 1,000 (3) 850 (4) 750